

Regular expressions

Regular expressions so far

Regular expression: a tool for specifying a search pattern in text.

Some regular expressions so far:

- `\d` any digit
- `+` one or more occurrences
- `^` anchors at the beginning
- `$` anchors at the end
- `\b` word boundary
- `|` alternation (this pattern OR that pattern)

Example: Cleaning phone numbers

You are working with customer data in which customers have entered their phone numbers:

```
[1] "555 867-5309"    "555 123 1234"    "(555) 298-9090"  "(555) 095 9876"  
[5] "5553246789"
```

You want to clean the numbers so they all have the same form. What would be the easiest approach?

Example: Cleaning phone numbers

```
1 str_remove_all(phone_nums, "\\D")
```

```
[1] "5558675309" "5551231234" "5552989090" "5550959876" "5553246789"
```

- `str_remove_all` removes all matches to a pattern
- `\d` matches any digit
- `\D` matches any *non*-digit

Shorthand character classes

- `\d` matches any digit
- `\w` matches any “word character” (letters, digits, underline)
- `\s` matches any “whitespace character” (space, tab, enter, new line)
- `\D`, `\W`, and `\S` are negations of `\d`, `\w`, and `\s`

Example 2: Selecting files

Here are a set of files that live on my computer:

```
1 file_names
```

```
[1] "research/project1/code.R"  
[2] "research/project1/data.csv"  
[3] "research/project2/sim_output.csv"  
[4] "teaching/sta279/lecture1.qmd"  
[5] "teaching/sta279/example_data.csv"  
[6] "teaching/sta279/research_project.html"
```

How would I select only the files that live in the *research* folder?

Example 2: Selecting files

Here are a set of files that live on my computer:

```
1 file_names
```

```
[1] "research/project1/code.R"  
[2] "research/project1/data.csv"  
[3] "research/project2/sim_output.csv"  
[4] "teaching/sta279/lecture1.qmd"  
[5] "teaching/sta279/example_data.csv"  
[6] "teaching/sta279/research_project.html"
```

How would I select only the files that live in the *research* folder?

```
1 str_subset(file_names, "^research")
```

```
[1] "research/project1/code.R"      "research/project1/data.csv"  
[3] "research/project2/sim_output.csv"
```

alternative: look for research /

Example 2: Selecting files

Here are a set of files that live on my computer:

```
1 file_names
```

```
[1] "research/project1/code.R"  
[2] "research/project1/data.csv"  
[3] "research/project2/sim_output.csv"  
[4] "teaching/sta279/lecture1.qmd"  
[5] "teaching/sta279/example_data.csv"  
[6] "teaching/sta279/research_project.html"
```

What about only `csv` files in the `research` folder?

· start with `research`

· end with `csv`

and anything is allowed in middle

`^research`  `csv$`
something

Example 2: Selecting files

Here are a set of files that live on my computer:

```
1 file_names
```

```
[1] "research/project1/code.R"
[2] "research/project1/data.csv"
[3] "research/project2/sim_output.csv"
[4] "teaching/sta279/lecture1.qmd"
[5] "teaching/sta279/example_data.csv"
[6] "teaching/sta279/research_project.html"
```

What about only csv files in the research folder?

```
1 str_subset(file_names, "^research.+\\.\\.csv$")
```

```
[1] "research/project1/data.csv"
"research/project2/sim_output.csv"
```

Handwritten notes:

- special character to match (basically)*
- anything*
- any string with one or more characters*
- literal period*

Dot

The `.` is a special character which matches (almost) any character:

```
1 str_view("I like bananas.", ".")
```

```
[1] | <I>< ><l><i><k><e>< ><b><a><n><a><n><a><s><.>
```

If we want to match a *literal* period, we need to escape it:

```
1 str_view("I like bananas.", "\\.")
```

```
[1] | I like bananas<.>
```

Example 3: Extracting LaTeX

LaTeX is a tool for scientific and mathematical typesetting.
For example:

```
$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$
```

becomes

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Example 3: Extracting LaTeX

Suppose we have a document which contains equations in LaTeX:

```
1 document_text
```

```
[1] "The equation for the simple linear regression line is given by  
$Y_i = \\beta_0 + \\beta_1 X_i + \\varepsilon_i$"
```

Question: If I want to extract only the equation, what pattern am I trying to match?

Example 3: Extracting LaTeX

```
1 document_text
```

```
[1] "The equation for the simple linear regression line is given by  
$Y_i = \\beta_0 + \\beta_1 X_i + \\varepsilon_i$"
```

```
1 str_extract(document_text, "\\$.+\\$")
```

```
[1] "$Y_i = \\beta_0 + \\beta_1 X_i + \\varepsilon_i$"
```

- Remember that `$` is a special character in regular expressions, meaning “the end of the string”. To get a *literal* dollar sign, we need the escape character: `\\$`

Example 3: Extracting LaTeX

```
1 document_text
```

```
[1] "The equation for the simple linear regression line is given by  
$Y_i = \\beta_0 + \\beta_1 X_i + \\varepsilon_i$"
```

```
1 str_extract(document_text, "(?<=\\$).+(?=\\$)")
```

```
[1] "Y_i = \\beta_0 + \\beta_1 X_i + \\varepsilon_i"
```

- `(?<=)` is a *positive lookbehind*. It is used to identify expressions which are *preceded* by a particular expression
- `(?=)` is a *positive lookahead*. It is used to identify expressions which are *followed* by a particular expression

Class activity

- Work independently or with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 15.4 - 15.7 in *R for Data Science*